

DIGITAL RESPIRATION RATE METER

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Measurement of physiological parameters like heart rate and respiration rate is crucial in the field of medicine. Advances in technology have provided different instruments for constantly monitoring these parameters. The commonly used transducers to measure respiration rate include thermistors placed in front of nostrils, micro-switches or displacement transducers put across the chest, impedance electrodes and

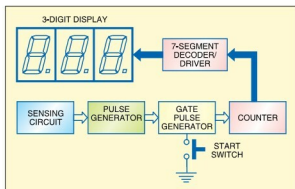


Fig. 1: Block diagram of the digital respiration rate meter

signals from CO₂ measurement.

Here is a simple method for respiration rate measurement using a displacement transducer. This meter can be used to monitor the respiration rate, pulse rate (by using a proper sensor) and heart rate. It responds fast and is cost-effective compared to conventional medical equipment. By using this, respiration rate can be measured in the range of 0-999 respirations/minute.

Circuit description

Fig. 1 shows the block diagram of the respiration rate meter. It uses a displacement transducer for sensing the respiration rate using IR transmitter and receiver as shown in the physical assembly. Inhaling and exhaling the air during respiration leads to movement of a lightweight ball (made of thermocol) up and down in a capillary glass tube. This movement is sensed with the help of IR transmitter-receiver assembly of the sensing circuit and converted into pulses through the pulse generator. These pulses are counted for a minute using a counter. The respiration rate is displayed on a 3-digit display through the 7-segment decoder/driver.

Start switch S1 is used to reset the

display to zero and enable the counter for a minute to count the respiration pulse. The gate pulse generator consists of a monostable multivibrator. When triggered by start switch, it generates gating pulse of one-minute duration.

Fig. 2 shows the circuit of the respiration rate meter. The IR transmitter LED (IRTX)

connected in series with resistor R1 transmits IR signals, which are received by the IR receiver LED (IRRX). The IR receiver is connected to the base of transistor T1 through resistor R2.

When the transmitted IR signal falls directly on the reverse-biased IR diode, it produces an electrical signal according to the IR intensity. So transistor T1 conducts and its collector goes low, which makes transistor T2 to cut off. Therefore the collector of transistor T2 becomes high, which represents logic '1'.

When the IR signal from the transmitter is interrupted due to movement of the ball up and down during the inhale-exhale mechanism, transistor T1 is cut-off and its collector goes high, which drives transistor T2 into conduction. The collector of transistor T2 goes low, which represents logic '0'. This means whenever the ball crosses the IR beam, a pulse is generated during inhale and exhale. Two pulses are generated due to the ball crossing the IR beam twice: one during inhale and the other during exhale. The final output of transistor T2 provides the clock pulse to IC 74LS90 (IC1).

IC 74LS90 is configured as a divide-by-two counter and its output

PARTS LIST

Semiconductors:

IC1	- 74LS90 decade counter
IC2	- NE555 timer
IC3	- CD4011 quad NAND gate
IC4	- CD4553 3-digit BCD counter
IC5	- CD4511 7-segment decoder/driver
IC6	- 7805, 5V regulator
T1, T2	- BC548 npn transistor
T3-T5	- BC558 pnp transistor
LED1	- 5mm LED
IRTX	- IR transmitting LED
IRRX	- IR receiving LED
D1-D5	- 1N4007 rectifier diode
DIS1-DIS3	- LTS543 common-cathode 7-segment display

Resistors (all 1/4-watt, ±5% carbon):

R1	- 100-ohm
R2, R4	- 1-kilo-ohm
R3, R5	- 10-kilo-ohm
R6	- 4.7-mega-ohm
R7-R9	- 4.7-kilo-ohm
R10-R16	- 270-ohm
R17	- 220-ohm
VR1	- 1-mega-ohm

Capacitors:

C1, C7	- 0.1µF ceramic disk
C2	- 10µF, 16V electrolytic
C3	- 0.01µF ceramic disk
C4	- 10nF ceramic disk
C5	- 1000µF, 25V electrolytic
C6	- 47µF, 16V electrolytic

Miscellaneous:

X1	- 230VAC primary to 9V, 500mA secondary transformer
S1	- Push-to-on switch